

STUDENT NUMBER: _____

TEACHER'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

Year 12

MATHEMATICS ADVANCED ASSESSMENT

HALF-YEARLY

March 2010

Time allowed – 3 hours + 5 minutes reading time

DIRECTIONS TO CANDIDATES:

- Start each question on a new page.
- Show all relevant working.
- Use black or blue pen.
- **NO** liquid paper is to be used.
- Approved Maths aids and calculators may be used.

QUESTION 1 [12 marks]**Marks**

- (a) Evaluate, correct to three significant figures

2

$$\frac{3.5^2}{1.8^2 - \sqrt{145}}$$

- (b) Find integers a and b such

2

$$(2\sqrt{5} - 1)^2 = a\sqrt{5} + b.$$

- (c) Solve $\frac{2x-1}{3} - \frac{1-3x}{5} = 2$

2

- (d) Find the primitive function of $3x + \sin 3x$

3

- (e) Find the value of x for which
 $|2x - 1| > 5$

2

- (f) Factorise $9x^2 - 16$

1**QUESTION 2** [12 marks]

- (a) Differentiate

(i) $x^2 \cdot \cos x$

2

(ii) $\frac{4-x}{\sin x}$

2

- (b) Solve $2 \cos x + 1 = 0$ for $0 \leq x \leq 2\pi$

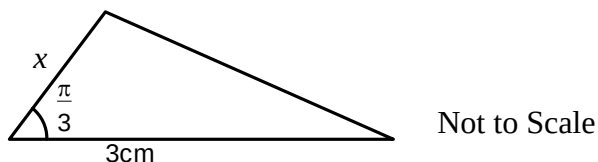
2

- (c) Evaluate

2

$$\int_{\frac{\pi}{12}}^{\frac{\pi}{9}} \sec^2 3x \, dx$$

- (d)



If the area of this triangle is 6 cm^2 , find the exact value of x .

3

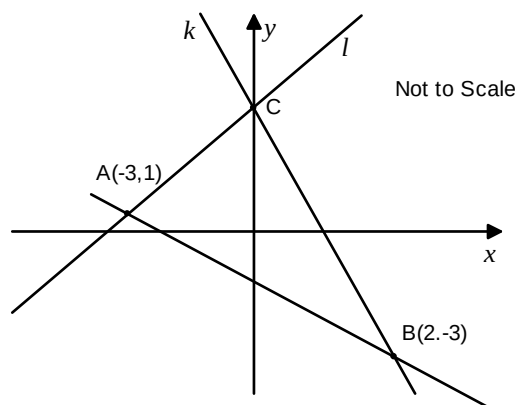
- (e) Sketch the curve of $y = \tan x$ for $0 \leq x \leq \pi$

1

QUESTION 3 [12 marks]

Marks

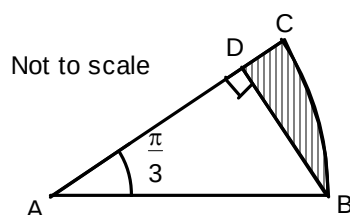
- (a) Given points $A(-3,1)$ and $B(2,-3)$.
 Point A lies on the line $l: 4x - 3y + 15 = 0$ and
 Point B lies on the line k given by the equation $4x + y - 5 = 0$.



- | | | |
|-------|---|---|
| (i) | Show that the point C , the point of intersection of the lines l and k , must lie on the y -axis. | 2 |
| (ii) | Find the gradient of the line AB | 1 |
| (iii) | Find the equation of AB | 2 |
| (iv) | Find the perpendicular distance from point C to the line AB | 2 |
| (v) | Find the area of $\triangle ABC$. | 2 |
- (b) Find the equation of the tangent to the curve $y = (x - 1)(x + 5)$ at the point where $x = 0$ 3

QUESTION 4 [12 marks]

(a)



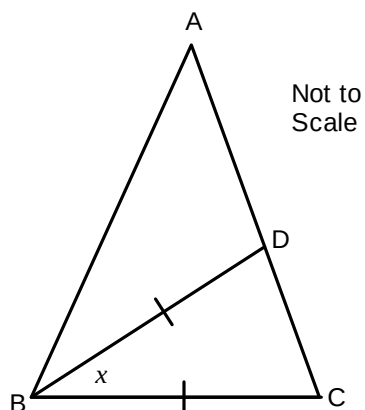
ABC is a sector with $\angle BAC = \frac{\pi}{3}$ and $AC = AB = 9\text{ cm}$

- | | | |
|------|---|---|
| (i) | Calculate the area of sector ABC | 1 |
| (ii) | Calculate the area of the shaded region | 3 |
- (b) Find the equation of the parabola with vertex $(1,4)$ and focus $(1,-2)$. 3
- (c) The tangent to the curve $y = \cos x$ at point P has a slope of $\frac{1}{2}$ for $0 \leq x \leq \frac{3\pi}{2}$. Find the coordinates of the point P . 3
- (d) Given $y = x^4 - x$. Is the function even or odd or neither? Justify your answer. 2

QUESTION 5 [12 marks]

Marks

(a)

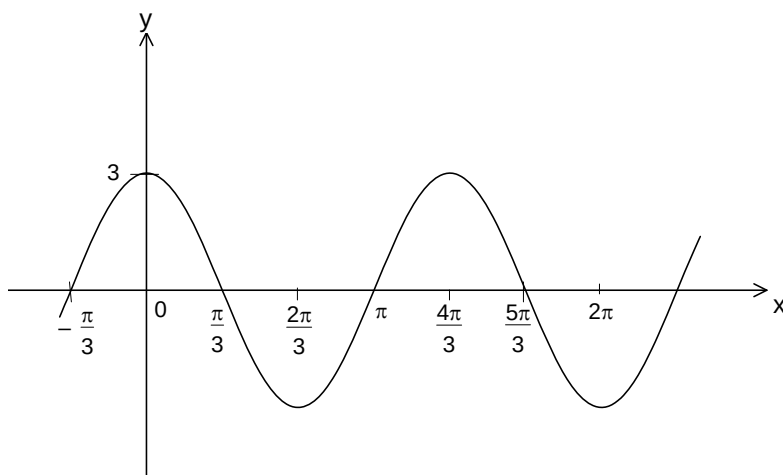


$\triangle ABC$ is isosceles where $AB = AC$.
 D lies on AC such that $\angle ABD = 3\angle DBC$ and also $BD = BC$.
 Find the value of x , giving reasons.

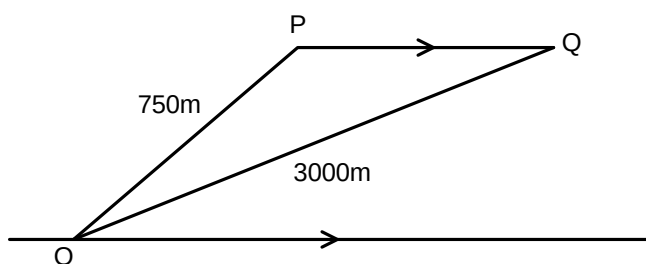
3

(b) The graph below can be represented by an equation in the form $y = a \cos nx$.
 Find the values of a and n .

2



(c)



An observer is standing at point O and sees a plane at P 750m from O .
 8 seconds later the plane is sighted at Q , 3000m from O .
 The angles of elevation of P and Q from O are 73° and 7° respectively.
 Find the speed of the plane.

4

(d) (i) Differentiate $y = (3x^2 - 7)^{10}$

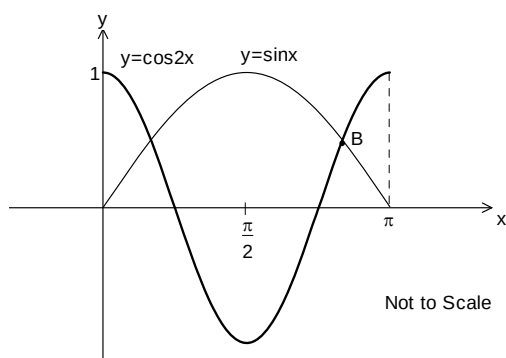
1

(ii) Hence find $\int x(3x^2 - 7)^9 dx$

2

QUESTION 6 [12 marks]**Marks**

- (a) (i) Graph 2
- $$y = \begin{cases} -\frac{3}{x-1} & \text{for } x < 0 \\ x^2 + 3 & \text{for } x \geq 0 \end{cases}$$
- (ii) Find $f(-2) + f(2)$ 1
- (iii) Find $f(a^2)$ 1
- (b) Find the values of k for which the quadratic equation $2x^2 - kx + k = 0$ has real roots 3
- (c) Use Simpson's rule with five function values to approximate the volume generated by $y = \sin x$ rotated around x -axis between $x = 0$ and $x = \frac{\pi}{2}$ 3
- (d) Find $\int 5 \cos x^\circ dx$ 2

QUESTION 7 [12 marks]**(a)**

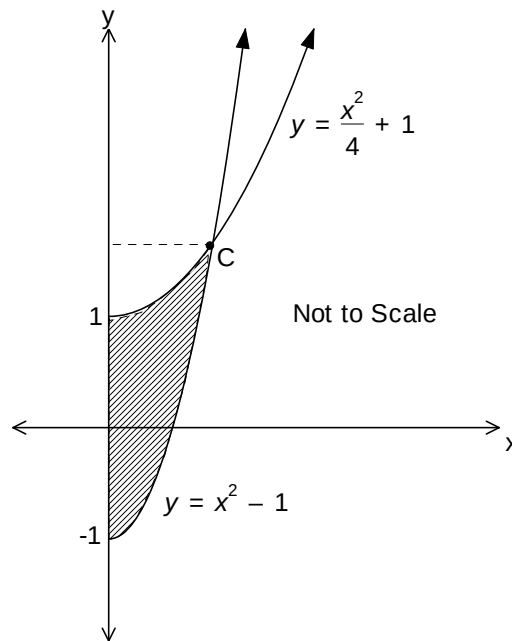
- (i) Show that $y = \cos 2x$ and $y = \sin x$ meet at the point where $x = \frac{\pi}{6}$ 2
Hence, write down the value of x at B .
- (ii) Redraw the diagram in your booklet and shade the region bounded only by these two curves in between $x = \frac{\pi}{6}$ and $x = \frac{\pi}{2}$ 1
- (iii) Find the exact area of the shaded region shown in part (ii) 3
- (b) The point $P(x, y)$ moves so its distance from the point $A(3, 9)$ is always twice the distance from the point $B(6, 6)$
- (i) Find the locus of the point P 2
- (ii) Show that the locus in part (i) is a circle. 2
State its centre and radius.
- (c) Evaluate 2
- $$\sum_{x=2}^{10} (3x - 5)$$

QUESTION 8 [12 marks]**Marks**

(a) Find the area bounded by $x = y^2 - 2y - 3$ and the y - axis

3

(b)



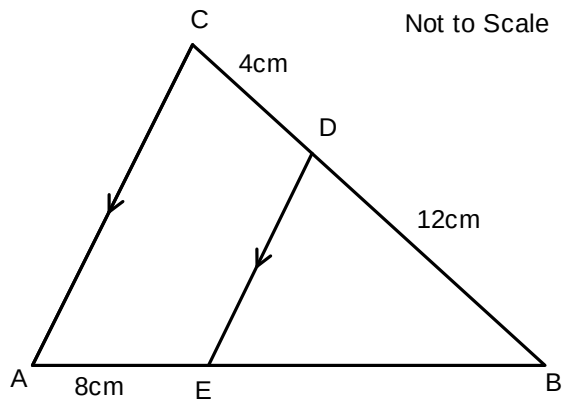
(i) Find the coordinates of C

2

(ii) Find the exact volume when the shaded area is rotated around the y - axis.

4

(c)



Given, $AC \parallel ED$, $CD = 4cm$, $DB = 12cm$ and $AE = 8cm$

(i) Prove that $\triangle CBA \sim \triangle DBE$

2

(ii) Hence or otherwise, calculate the length of EB

1

QUESTION 9 [12 marks]**Marks**

- (a) (i) Prove that $\sec^2 \theta - 2 \tan \theta = (\tan \theta - 1)^2$ 2
- (ii) Hence or otherwise solve
 $\sec^2 \theta - 2 \tan \theta = 0$ for $0 \leq \theta \leq 2\pi$ 2
- (b) Consider the geometric series
 $3 - 6x + 12x^2 - 24x^3 + \dots$
- (i) For what values of x does this series have a limiting sum? 2
- (ii) If the limiting sum of the series is 2.5, find the value of x 2
- (c) The parabola is given by equation $2y = x^2 - 8x$
Find:
- (i) The coordinates of the vertex 1
- (ii) The focal length 1
- (iii) The focus 1
- (iv) The equation of the directrix 1

QUESTION 10 [12 marks] **Marks**

(a) Given that $f(x) = x^2\sqrt{10-x}$

(i) Show that $f'(x) = \frac{5x(8-x)}{2\sqrt{10-x}}$

2

(ii) State the domain of the function

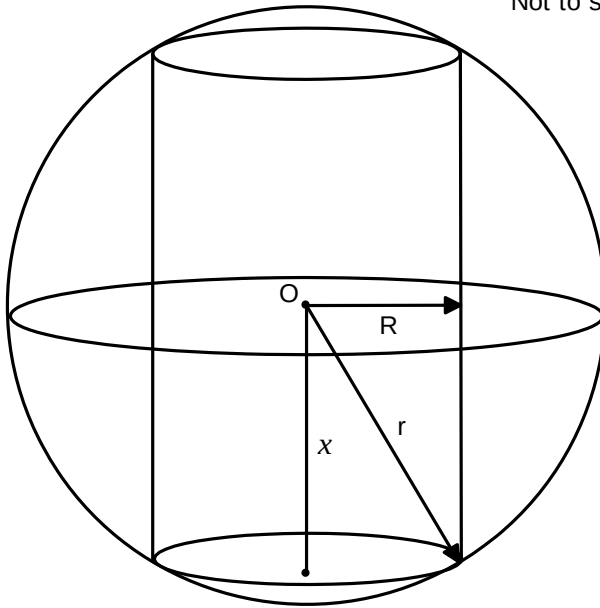
1

(iii) Find all the stationary points on the curve $y = x^2\sqrt{10-x}$ and determine their nature.

3

(b) A cylinder is to be made to fit inside a sphere of radius r cm, as shown

Not to scale



(c)

Let R = radius of cylinder and x be the distance of the base of the cylinder from the centre of the sphere as shown.

(i) Find an expression for the radius of the cylinder (R) in terms of r and x

1

(ii) Show that the volume, V , of the cylinder is given by $V = 2\pi x(r^2 - x^2)$

2

(iii) Find in terms of r , the maximum volume of the cylinder. Leave your answer in exact form.

3

Question 1

a) $-1.39179... = -1.39$ ✓

b) $(2\sqrt{5}-1)^2 = a\sqrt{5} + b$
 $4 \times 5 - 2 \times 2\sqrt{5} + 1 = a\sqrt{5} + b$
 $20 + 1 - 4\sqrt{5} = b + a\sqrt{5}$
 $b = 21 \quad a = -4$ ✓

c) $\frac{2x-1}{3} - \frac{1-3x}{5} = 2$
 $\frac{10x-5-3+9x}{15} = 2$ ✓
 $19x = 30+8$
 $\therefore x = 2$ ✓

d) $\int 3x + \sin 3x \, dx$
 $= \frac{3x^2}{2} - \frac{1}{3} \cos 3x + C$ ✓

e) $|2x-1| > 5$
 $2x-1 > 5 \quad \text{or} \quad 2x-1 < -5$
 $x > 3 \quad \checkmark \quad x < -2$
 $\therefore x < -2, x > 3$ ✓

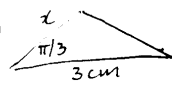
f) $(3x-4)(3x+4)$ ✓

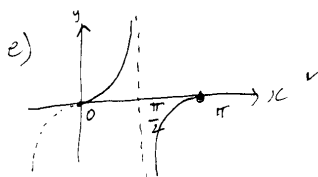
Question 2 O.K.

a) $\frac{d}{dx} (x^2 \cdot \cos x)$
 i) $\frac{d}{dx} = 2x \cdot \cos x + x^2 \cdot (-\sin x)$
 ii) $\frac{d}{dx} \left(\frac{4-x}{\sin x} \right) = \frac{-1 \cdot \sin x - (4-x) \cdot \cos x}{\sin^2 x}$
 $= \frac{-\sin x - 4\cos x + x \cdot \cos x}{\sin^2 x}$ ✓

b) $2\cos x + 1 = 0$
 $\cos x = -\frac{1}{2}$
 $x = \frac{2\pi}{3}, \frac{4\pi}{3}$ ✓

c) $\int_{\pi/12}^{\pi/4} \sec^2 3x \, dx = \left[\frac{1}{3} \tan 3x \right]_{\pi/12}^{\pi/4}$
 $= \frac{1}{3} \left[\tan \frac{\pi}{3} - \tan \frac{\pi}{4} \right] = \frac{1}{3} [\sqrt{3} - 1]$ ✓ or 0.244 ✓

d) 
 $A = 6 = \frac{1}{2} x \times 3 \times \sin \frac{\pi}{3}$ ✓
 $12 = 3x \cdot \frac{\sqrt{3}}{2}$
 $x = \frac{24}{3\sqrt{3}} = \frac{8}{\sqrt{3}}$ ✓



Question 3.

a) $A(-3, 1) \quad B(2, -3)$
 $\ell: 4x-3y+15=0 \quad k: 4x+y-5=0$

i) $C \begin{cases} 4x-3y+15=0 \\ 4x+y-5=0 \end{cases} \therefore 4y-20=0 \quad y=5$

$\therefore y=5 \quad 4x-3y+15=0$
 $4x-15+15=0 \quad (x=0)$
 $\therefore C(0, 5) \therefore \text{lies on } y\text{-axis}$

ii) $m_{AB} = \frac{1+3}{-3-2} = -\frac{4}{5}$ ✓

iii) $AB: y-1 = -\frac{4}{5}(x+3)$ ✓
 $y = -\frac{4}{5}x - \frac{7}{5}$ ✓

or $4x+5y+7=0$

iv) $C(0, 5) \quad AB: 4x+5y+7=0$
 $\therefore d = \frac{|4 \times 0 + 5 \times 5 + 7|}{\sqrt{4^2+5^2}} = \frac{32}{\sqrt{41}}$ ✓

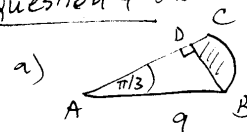
v) $\text{Area} = \frac{1}{2} \times AB \times \frac{1}{2} \quad AB = \sqrt{41}$
 $= \frac{1}{2} \times \sqrt{41} \times \frac{32}{\sqrt{41}}$ ✓
 $= 16 \text{ (units}^2\text{)}$ ✓

b) $y = (x-1)(x+5) = x^2 + 4x - 5$
 $\therefore \frac{dy}{dx} = 2x+4 \quad \therefore m = 2 \times 0 + 4 = 4$

at $x=0 \quad y=-5$ ✓

$\therefore \text{tangent: } y+5 = 4(x-0)$ ✓
 $y = 4x-5$

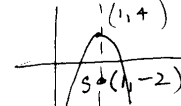
Question 4 O.K.



i) $\text{Area} = \frac{1}{2} r^2 \theta = \frac{1}{2} \times \dots$
 $\therefore A = \frac{27\pi}{2}$ ✓

ii) $\triangle ADB = \frac{1}{2} AD \cdot AB$
 $\cos \frac{\pi}{3} = \frac{AD}{9} = \frac{1}{2} \therefore AD = \frac{9}{2}$
 $\therefore \text{Area} = \frac{1}{2} \times \frac{9}{2} \times 9 \times \frac{\sqrt{3}}{2}$
 $= \frac{81\sqrt{3}}{8}$ ✓

$\therefore \text{Area} = \frac{27\pi}{2} - \frac{81\sqrt{3}}{8}$
 or $= \frac{108\pi - 81\sqrt{3}}{8}$

b) $(1, 1) \quad (1, -2)$


$(x-m)^2 = -4a(y-m)$
 $(x-1)^2 = -24(y+0.5)$ ✓

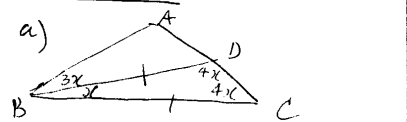
c) $y = \cos x$
 $\frac{dy}{dx} = -\sin x \quad P(x) = \dots$
 $\frac{1}{2} = -\sin x$ ✓
 $\therefore x = \frac{7\pi}{6}$ (only)

$\therefore P\left(\frac{7\pi}{6}, -\frac{\sqrt{3}}{2}\right)$

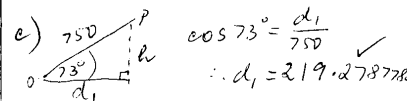
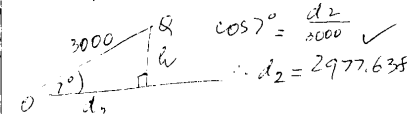
Question 4d)

$y = x^4 - x = f(x)$
 $f(-x) = (-x)^4 - (-x) = x^4 + x$
 $\neq f(x) \therefore$ not even
 $f(-x) \neq -f(x)$
 since $f(-x) = x^4 + x = -(x^4 - x)$
 $\neq -f(x) \therefore$ not odd
 $\therefore$ neither

Question 5 v.k.

a) 
 $\angle ABD = 3x \therefore \angle ABC = 4x$
 $\therefore \angle ACB = 4x$ (base angles of an isosceles triangle)
 $\therefore \angle BDC = 4x$ since $BD = DC$
 $\therefore x + 4x + 4x = 180^\circ$ (angle sum of $\triangle$)
 $\therefore x = 20^\circ$

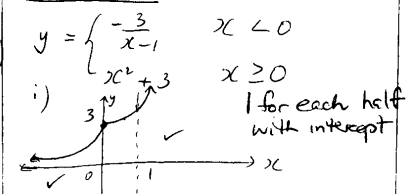
b) $y = a \cos nx$
 $a = 3$ (amplitude)
 $T = \frac{4\pi}{3} = \frac{2\pi}{n} \therefore n = \frac{3}{2}$

c) 
 $\cos 73^\circ = \frac{d_1}{750}$
 $\therefore d_1 = 219.278728$

 $\cos 7^\circ = \frac{d_2}{3000}$
 $\therefore d_2 = 2977.638455$

$\therefore d_{\text{TRAVELLED}} = d_2 - d_1 = 2758.3597$
 $\therefore \text{speed} = \frac{2758.35976m}{8 \text{ sec.}}$
 $= 344.7949596 \text{ m/s}$

d) i) $y = (3x^2 - 7)^{10}$
 $\frac{dy}{dx} = 10(3x^2 - 7)^9 \times 6x$
 $= 60x(3x^2 - 7)^9$
 ii) $\therefore y = \int 60x(3x^2 - 7)^9 dx$
 $\therefore \frac{y}{60} = \int x(3x^2 - 7)^9 dx$
 $\therefore \int x(3x^2 - 7)^9 dx = \frac{1}{60} (3x^2 - 7)^{10} + C$

Question 6 v.k.

$y = \begin{cases} -\frac{3}{x-1} & x < 0 \\ x^2 + 3 & x \geq 0 \end{cases}$
 i) 
 ii) $f(-2) + f(2) = \frac{-3}{-2-1} + 2^2 + 3$
 $= \frac{-3}{-3} + 4 + 3 = 1 + 4 + 3 = 8$
 iii) $f(a^2) = (a^2)^2 + 3 = a^4 + 3$

Question 6

b) $2x^2 - kx + k = 0$
 $\Delta = b^2 - 4ac \geq 0$
 $(-k)^2 - 4 \times 2 \times k \geq 0$
 $k^2 - 8k \geq 0$
 $k(k - 8) \geq 0$
 $\therefore k \leq 0, k \geq 8$

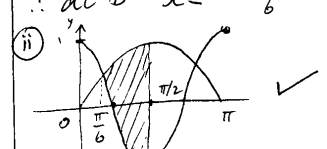
c) $y = \sin x$
 $V = \pi \int_{\pi/2}^{\pi} \sin x dx$

x	0	$\pi/8$	$\pi/4$	$3\pi/8$	$\pi/2$
$\sin x$	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	1

 $V = \pi \times \frac{\pi}{8} \left(0 + 1 + 4 \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) \right)$
 $+ 2 \times \sin \frac{\pi}{4} = \frac{\pi}{4} \times 6 = \frac{3\pi}{2}$
 1 mark off each error 24
 1 mark for correct area answer 1
 or $V = 2.4674$

d) $\int 5 \cos x^\circ dx = \int 5 \cos \frac{x\pi}{180} dx$
 $= 5 \times \frac{180}{\pi} \cdot \sin \frac{x\pi}{180} + C$
 or $5 \times \frac{180}{\pi} \cdot \sin x^\circ + C$

Question 7 v.k.

a) $y = \cos 2x$ and $y = \sin x$
 $x = \frac{\pi}{6} \therefore y = \cos \frac{\pi}{3} = \frac{1}{2}$ $y = \sin \frac{\pi}{6} = \frac{1}{2}$
 $\therefore$ at $x = \frac{\pi}{6}$ pt. of intersection
 $\therefore$ at B $x = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$
 ii) 
 iii) $A = \int_{\pi/6}^{5\pi/6} \sin x - \cos 2x dx$
 $= \left[-\cos x - \frac{1}{2} \sin 2x \right]_{\pi/6}^{5\pi/6}$
 $= -\cos \frac{5\pi}{6} - \frac{1}{2} \sin \pi + \cos \frac{\pi}{6} + \frac{1}{2} \sin \frac{\pi}{3}$
 $= 0 - 0 + \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{\sqrt{3}}{2}$
 $= \frac{3}{4} \sqrt{3}$
 b) $PA = 2 PB$ A(3, 6) B(6, 6)
 $\sqrt{(x-3)^2 + (y-6)^2} = 2\sqrt{(x-6)^2 + (y-6)^2}$
 $x^2 - 6x + 9 + y^2 - 12y + 36 = 4(x^2 - 12x + 36 + y^2 - 12y + 36)$
 $9 + 36 - 4x^2 + 24x - 36 - 4y^2 + 48y - 144 = 0$
 $-4x^2 + 24x - 4y^2 + 48y - 105 = 0$
 $x^2 - 6x + y^2 - 12y - \frac{105}{4} = 0$
 $8 = (x-7)^2 + (y-5)^2$
 ii) $8 = (x-7)^2 + (y-5)^2$
 $\therefore$ centre (7, 5)
 radius $r = \sqrt{8}$

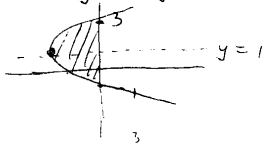
Question 7 c)

$$\sum_{x=2}^{10} 3x-5 = 1+4+7+\dots+25$$

$$x=2 \quad \begin{aligned} &= \frac{n}{2}(a+l) \quad \begin{matrix} A.P. \\ n=9 \\ a=1 \quad l=25 \end{matrix} \\ &= \frac{9}{2}(1+25) = 117 \checkmark \end{aligned}$$

Q.8 o.k.

a) $x = y^2 - 2y - 3$ y -axis
 $x = (y-3)(y+1)$



$$\therefore A = \left| \int_{-1}^3 (y^2 - 2y - 3) dy \right| \checkmark$$

$$= \left| \left[\frac{y^3}{3} - y^2 - 3y \right]_{-1}^3 \right| = \frac{32}{3} \checkmark$$

b) $\begin{cases} y = \frac{x^2}{4} + 1 \\ y = x^2 - 1 \end{cases} \quad \frac{x^2}{4} + 1 = x^2 - 1$
 $\therefore 8 = 3x^2 \quad x^2 = \frac{8}{3}$
 $\therefore x = \pm \sqrt{\frac{8}{3}}$
 $y = x^2 - 1 = \frac{8}{3} - 1 = \frac{5}{3}$
 $\therefore C \left(\pm \sqrt{\frac{8}{3}}, \frac{5}{3} \right)$

ii) $V_1 = \pi \int_{-1}^{5/3} (y+1) dy$ $y = x^2 - 1$
 $V_2 = \pi \int_{-1}^{5/3} (4y-4) dy$ $y = \frac{x^2}{4} + 1$
 $\therefore V = V_1 - V_2$

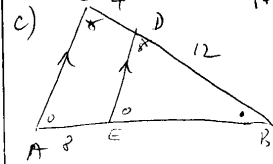
$$\therefore V = \pi \int_{-1}^{5/3} (y+1) dy - \pi \int_{-1}^{5/3} (4y-4) dy$$

$$= \pi \left[\frac{y^2}{2} + y \right]_{-1}^{5/3} - \pi \left[2y^2 - 4y \right]_{-1}^{5/3}$$

$$= \pi \left[\frac{25}{18} + \frac{1}{2} \right] - \pi \left[-\frac{10}{9} - (-2) \right]$$

$$= \frac{32}{9}\pi - \frac{8}{9}\pi = \frac{8}{3}\pi \checkmark$$

if not splitted $\therefore \frac{3}{4}$



i) $\angle CBA = \angle DBA$ (in common)
 $\angle BCA = \angle BDE$ $\checkmark$ (corresp. $\angle$ s)
 $\angle DEB = \angle CAB$ $\checkmark$ (on parallel lines)
 $\therefore \triangle CBA \sim \triangle DBE$ (all angles are $\sim$)

ii) $\frac{EB}{AB} = \frac{DB}{CB}$ all similar sides are in the same ratio
 $\therefore \frac{EB}{8+EB} = \frac{12}{12+4}$
 $16 \times EB = 12(8+EB)$
 $4EB = 96$
 $\therefore EB = 24 \checkmark$

Question 9 o.k.

a) i) RHS = $(\tan \theta - 1)^2$
 $= \tan^2 \theta - 2 \tan \theta + 1$
 $= \sec^2 \theta - 2 \tan \theta$ $\checkmark$
 $= \sec^2 \theta - 2 \tan \theta = \text{LHS}$ $\checkmark$ *proven*

ii) $\sec^2 \theta - 2 \tan \theta = 0$ $0 \leq \theta < 2\pi$
 $\therefore (\tan \theta - 1)^2 = 0$
 $\tan \theta = 1 \therefore \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

b) $3 - 6x + 12x^2 - 2 + x^3 + \dots$
 i) $r = -\frac{6x}{3} = -2x \checkmark$

lim. sum exists $|r| < 1 \therefore |-2x| < 1$
 $\checkmark \quad \begin{cases} |x| < \frac{1}{2} \\ -\frac{1}{2} < x < \frac{1}{2} \end{cases}$

ii) $S_{\infty} = \frac{a}{1-r} = \frac{3}{1-(-2x)} = 2.5$
 $\therefore x = \frac{1}{10} \checkmark$

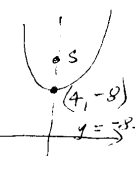
c) $2y = x^2 - 8x$

i) $2y + 16 = x^2 - 8x + 16$
 $2(y+8) = (x-4)^2$
 $\therefore \text{vertex } (4, -8) \checkmark$

ii) $4a = 2 \therefore a = \frac{1}{2} \checkmark$

iii) $S(4, -7\frac{1}{2}) \checkmark$

iv) $y = -8\frac{1}{2} \checkmark$



Question 10 o.k.

a) $f(x) = x^2 \sqrt{10-x} = x^2 (10-x)^{1/2}$
 i) $f'(x) = 2x \cdot (10-x)^{1/2} + x^2 \cdot \frac{1}{2} (10-x)^{-1/2}$

$$\therefore f'(x) = \frac{2x(10-x)^{1/2}}{1} + \frac{x^2}{2(10-x)^{1/2}}$$

$$= \frac{2x2x(10-x) - x^2}{2\sqrt{10-x}} \checkmark$$

$$\therefore f'(x) = \frac{5x(8-x)}{2\sqrt{10-x}}$$

ii) $10-x \geq 0 \therefore x \leq 10$

iii) $f'(x) = 0 = \frac{5x(8-x)}{2\sqrt{10-x}}$
 $\therefore 0 = 5x(8-x)$
 $x = 0 \quad \text{or} \quad x = 8$

values $\begin{matrix} x & -1 & 0 & 1 \\ f'(x) & \frac{5}{2\sqrt{11}} & 0 & \frac{35}{2\sqrt{9}} \end{matrix} \quad \begin{matrix} \sqrt{\frac{x}{10-x}} \\ f(x) & \frac{5}{2\sqrt{11}} & 0 & \frac{35}{2\sqrt{9}} \end{matrix}$

$\therefore (0, 0) \checkmark$ (Min. turning)
 $(8, 64) \checkmark$ (Max. turning)

Question 10 cont.

b) i) $R^2 = r^2 - x^2$
 $\therefore R = \downarrow \sqrt{r^2 - x^2}$ ✓
only

ii) $V = \pi \times R^2 \times h$ ✓ $h = 2x$

$\therefore V = \pi \times (r^2 - x^2) \times 2x$ ✓

$\therefore V = 2\pi x (r^2 - x^2)$

iii) $\frac{dV}{dx} = 0$

$V = 2\pi r^2 x - 2\pi x^3$

$\therefore \frac{dV}{dx} = 2\pi r^2 - 6\pi x^2$ ✓

$0 = 2\pi r^2 - 6\pi x^2$

$x^2 = \frac{2\pi r^2}{6\pi} = \frac{r^2}{3} \therefore x = \frac{+r}{\sqrt{3}}$ ✓

$\therefore \frac{d^2V}{dx^2} = 0 - 12\pi x$

at $x = \frac{r}{\sqrt{3}} \therefore \frac{d^2V}{dx^2} = -12\pi x \frac{r}{\sqrt{3}} < 0$

$\therefore$ at $x = \frac{r}{\sqrt{3}}$ V is Max.

$\therefore V_{MAX} = 2\pi \overbrace{x}^{\frac{r}{\sqrt{3}}} (r^2 - \underbrace{x^2}_{\frac{r^2}{3}})$

$\therefore V_{MAX} = \frac{2\pi r}{\sqrt{3}} \left(r^2 - \frac{r^2}{3} \right)$ ✓

(or) $V_{MAX} = \frac{2\pi r}{\sqrt{3}} \left(\frac{2r^2}{3} \right)$
 $= \frac{4\pi r^3}{3\sqrt{3}}$